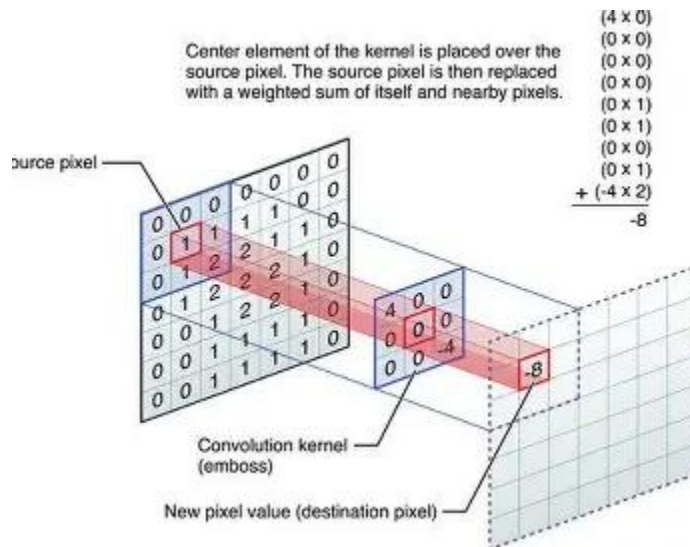
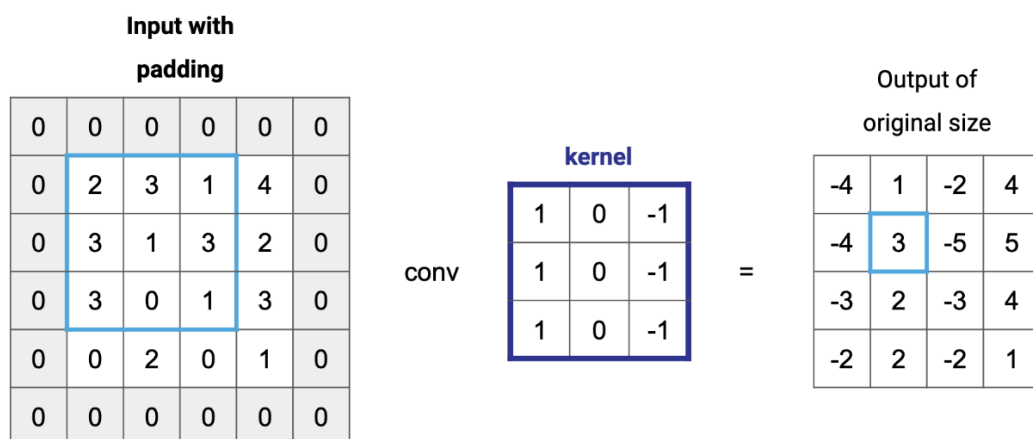
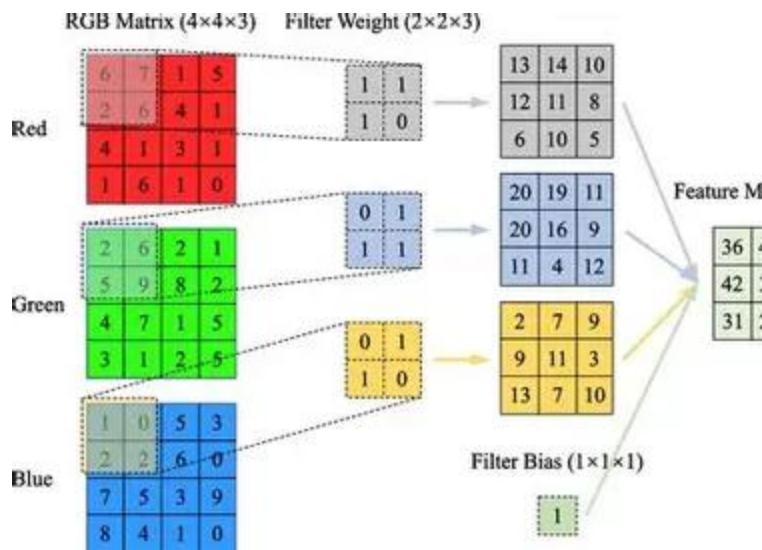


Convolution is an important operation in signal and image processing. Convolution operates on two signals (in 1D) or two images (in 2D): you can think of one as the input signal (or image), and the other (called the kernel) as a filter on the input image, producing an output image (so convolution takes two images as input and produces a third as output). Convolution is an incredibly important concept in many areas of math and engineering (including computer vision, as well see later).

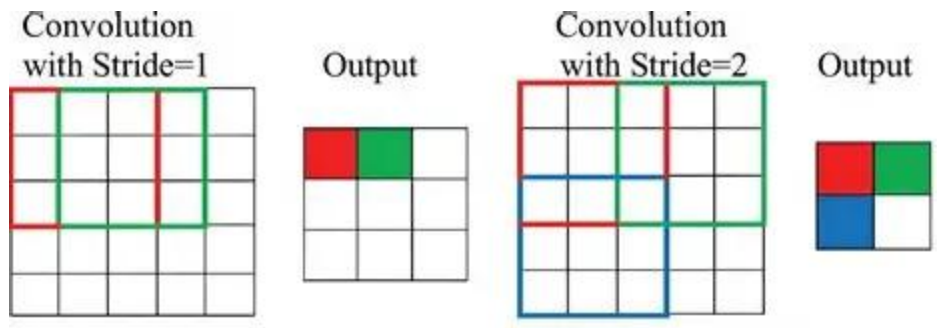
You can visually see the operation here [Visualizing convolution operation](#)

You observe that a convolution operation is a sum of the element wise multiplication operations on a window of the image where the filter is placed. The kernel is slid across the image to calculate the output. A filter of $k \times k$ on an image of $m \times n$ will give an output image of $(m - k/2) \times (n - k/2)$. To get a $m \times n$ output, the input has to be padded with $(k - k/2)$ on all sides ,i.e. top row, bottom row, left column and right column. The padding can be with 0 or boundary pixels. A few sample examples are given below:

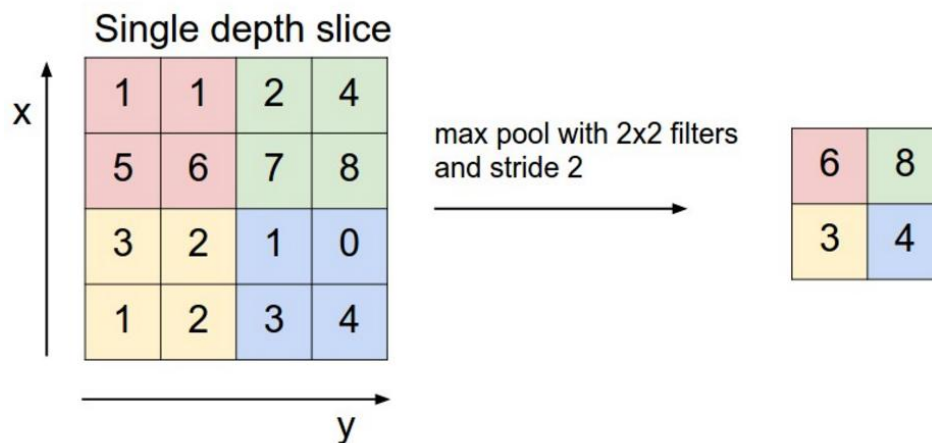




Note that square odd filters i.e. 3×3 , 5×5 are more common than rectangular even filters i.e. 2×4 . Odd filters are more suitable for centering while square filters give same weight to all neighbours. When using the `nn.Conv2d` function you need to mention no. of input channels, no. of filters to be applied, kernel size, padding and stride. Stride denotes the slide step of the window, whether moved by 1 pixel or more. A bigger stride is used when we do not want features from closer regions. This means after computing features (via convolution filter) from one window, we can skip a few pixels and then compute features again. This is when the image is very big and neighboring areas are repetitive, so we don't miss much with a bigger stride. The default value for padding is 0 and stride is 1. An example of stride is shown below:



There is another concept of dilation which we will discuss later. Lets see some examples of maxpooling and linear layers next.



An example of linear layers is given below. We translate 1×4 input to 1×3 using a 4×3 weight matrix. Then a 3×2 weight matrix is used to translate 1×3 to 1×2 .

